YEAR 9 MATHS · CHAPTER 3 · MIND MAP

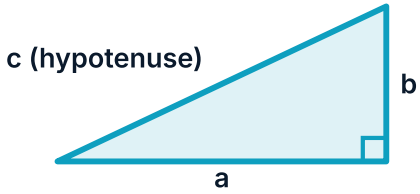
Pythagoras and trigonometry

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1 Pythagoras' theorem

In a right-angled triangle: $c^2 = a^2 + b^2$. c is the hypotenuse, the longest side.

Sides 6 and 8: $c^2 = 36 + 64 = 100$, so $c = 10$



2 Finding a shorter side

To find a shorter side, subtract: $a^2 = c^2 - b^2$.

Hypotenuse 13, side 5: $a^2 = 169 - 25 = 144$, so $a = 12$

Watch out: a shorter side can never be longer than the hypotenuse.

3 Pythagoras in 2D problems

Draw it, mark the right angle, then find the triangle.

A 5 m ladder reaches 4 m up a wall: the foot is $\sqrt{25 - 16} = 3$ m out

Diagonal of a 9 by 12 rectangle: $\sqrt{81 + 144} = 15$

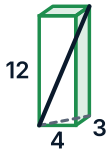
4 Pythagoras in 3D

Find a right triangle on one face first, then use that length in a second triangle.

Box 3 by 4 by 12: base diagonal $\sqrt{9 + 16} = 5$

Space diagonal $\sqrt{5^2 + 12^2} = 13$

Shortcut: $d = \sqrt{a^2 + b^2 + c^2}$

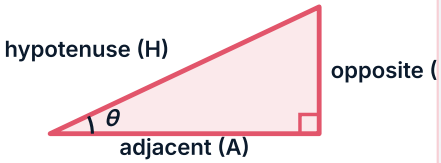


Floor diagonal 5
Space diagonal 13

5 Trigonometric ratios

SOH CAH TOA. Label the sides from the angle you're using.

$\sin\theta = \frac{O}{H}$ $\cos\theta = \frac{A}{H}$ $\tan\theta = \frac{O}{A}$



Watch out: set your calculator to degrees.

6 Finding unknown sides

Pick the ratio that uses the side you know and the side you want.

Opposite 35°, hypotenuse 10: $x = 10\sin 35° \approx 5.74$

Adjacent 8, angle 40°, find opposite: $x = 8\tan 40° \approx 6.71$

7 Solving for the denominator

When x is on the bottom, swap it with the trig value.

$\cos 30° = \frac{12}{x} \Rightarrow x = \frac{12}{\cos 30°} \approx 13.86$

Watch out: check the answer makes sense: the hypotenuse must be the longest side.

8 Finding unknown angles

Use the inverse buttons: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.

Opposite 5, hypotenuse 10: $\sin\theta = 0.5$, so $\theta = 30°$

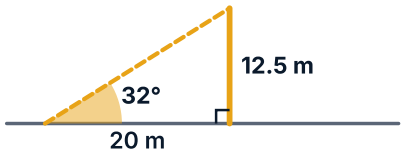
Opposite 7, adjacent 4: $\theta = \tan^{-1}(1.75) \approx 60.3°$

9 Applying trigonometry

Angles of elevation (looking up) and depression (looking down) are measured from the horizontal.

20 m from a tree, the angle of elevation to the top is 32°

Height $= 20\tan 32° \approx 12.5$ m



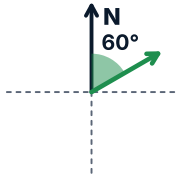
Watch out: draw the horizontal line first, then the angle.

10 Bearings

True bearings are measured clockwise from north, with three digits.

North-east is 045°T, south-east 135°T, west 270°T

Walk 10 km on 060°T: east $10\sin 60° \approx 8.66$ km, north $10\cos 60° = 5$ km



060°T
clockwise
from north